

- **Research field of the orientative paper : Biomedical signal acquisition systems using surface electromyography (sEMG) for monitoring muscle activity, gesture recognition, motor rehabilitation, and human-machine interface control.**

Problem addressed:

Accurate muscle activity monitoring is essential in assistive and rehabilitation systems, yet traditional Ag/AgCl gel electrodes are limited by discomfort, motion artefacts, and short-duration usability.

Sensor techniques analyzed:

- Ag/AgCl gel electrodes (reference paper) — standard clinical quality but not suitable for daily wear.
- Soft screen-printed flexible electrode arrays (*Luxembourg et al., 2025*) — enable continuous joint-angle estimation and gesture recognition during natural arm motion.
- Textile-integrated dry electrodes (*Günther et al., 2024*) — designed for long-term monitoring, embedded into clothing.
- Multi-channel EMG-based gesture recognition for amputees (*IEEE, 2019*) — supports intuitive prosthetic hand control.
- EMG acquisition & processing review (*Gohel & Mehendale, 2020*) — explains how sensor type, placement, filtering, and feature extraction determine signal quality.

Measurement principle:

Muscle contraction produces action potentials detectable at the skin surface as microvolt-level biopotentials.

Signal acquisition chain:

Electrodes → Instrumentation amplifier (high CMRR) → Band-pass filtering ($\approx 20\text{--}450$ Hz) → ADC → Feature extraction → Machine-learning classifier / regression model.

Key findings across studies:

- Flexible printed arrays: $R^2 \approx 0.85$ for joint-angle estimation and $\sim 87\%$ accuracy in gesture recognition.
- Textile electrodes: Comparable signal quality to Ag/AgCl while improving comfort and long-term wearability.
- Amputee gesture systems: Multi-channel EMG + ML allows intuitive prosthetic control, confirming practical feasibility.

Applications:

Rehabilitation monitoring • Prosthetic control • Wearable movement interfaces • Daily-life muscle tracking

- **Discussion on the topic**

The papers collectively show a shift from gel-based clinical electrodes to flexible, wearable, and textile-integrated EMG sensors that work under natural movement conditions. When paired with proper signal conditioning and machine-learning models, these systems can provide reliable gesture recognition and motion estimation for rehabilitation and prosthetic control. However, signal variability, motion-related noise, and the need for individual calibration remain common challenges. Even so, current progress strongly supports the development of practical, comfortable, and daily-use EMG devices.

Contribution of the members of the team in the project

Grade each item between 5 and 0 in the next tables. 5 means very well, 0 means very bad.

Rate the contribution of other members to the project. - Overall Group E

Yuxi Qiao	5	Participated actively in the group discussion and focused on summarizing the evolution of EMG electrode materials from clinical to textile-based systems. Contributed to structuring the final summary slide and ensuring clarity in the technical terminology.
Kamela Xhengo	5	Compared the selected papers by pinpointing shared technical limitations, particularly calibration and signal noise issues. Integrated these insights into the individual report and helped consolidate the group's final conclusions, ensuring the presentation was clear, coherent, and well-aligned across members.
Marco Russo	5	After reading the papers, I prepared the summary of the sensor techniques used in each study and explained them to the other group members. I also contributed to building the final presentation and refining the slide content.
Carmen Aznar Mathonneau	5	Collected data from the assigned papers and focused on analyzing the machine learning approaches applied to EMG signal decoding. Helped revise the visual design and consistency of the final summary slide.